

Review of Power Series

in this section we will be dealing with series we mostly will be dealing with ratio test:

review on ratio:

$$\lim_{n \rightarrow \infty} \left| \frac{C_{n+1}}{C_n} \right| |x - a| = L$$

if $L < 1$ = converges

if $L > 1$ = diverges

Radius of convergence

this tells you how far from the center can x go before the series breaks

Practice Problems

#1)
$$\sum_{n=1}^{\infty} \frac{(-1)^n}{n} x^n$$

$$C_{n+1} = \frac{(-1)^{n+1}}{n} x^{n+1}$$

• for this all you are doing is add a one to every exponent n

$$C_n = \frac{(-1)^n}{n} x^n$$

• this just stays the same



• this has to be reciprocal

$$\lim_{n \rightarrow \infty} \left| \frac{(-1)^{n+1}}{n} x^{n+1} \cdot \frac{n}{(-1)^n x^n} \right| \rightarrow \left| \frac{\cancel{(-1)^n} (-1)^1 \cancel{x^n} x^1}{\cancel{n}} \cdot \frac{\cancel{n}}{\cancel{(-1)^n} x^n} \right|$$

$$\lim_{n \rightarrow \infty} (-1)^n x \Rightarrow |x| < 1$$

• you must include the absolute value

So now we can find R which is one we got R From $|x| < 1$
 $\uparrow$
 R

We can now get rid of the absolute value by doing this:

$-1 < x < 1$ the plug in each x value into the original eqn

$$@ x = -1 \Rightarrow \sum_{n=1}^{\infty} \frac{(-1)^n}{n} (-1)^n \Rightarrow \sum_{n=1}^{\infty} \frac{1}{n}$$

• this is a harmonic series it will always diverge

harmonic series

a harmonic series looks like this:

$$\sum_{n=0}^{\infty} \frac{1}{n}$$

it will always diverge meaning if you ever have a problem that looks like this it will diverge

$$@ x = 1 \rightarrow \sum_{n=1}^{\infty} \frac{(-1)^n}{n} (1)^n \Rightarrow \sum_{n=1}^{\infty} \frac{(-1)^n}{n}$$

• this is a alternating series

$$b_n = \frac{1}{n} \rightarrow \lim_{n \rightarrow \infty} \frac{1}{n} = 0$$

$$\frac{d}{dn}(b_n) \rightarrow \frac{d}{dn}\left(\frac{1}{n}\right) = -\frac{1}{n^2}$$

} the limit is equal to zero and its decreasing so it converges

- use "C" if diverge
- use "L" if converges

Interval: $[-1, 1]$
 R: 1

• begin with smaller #

alternating series

an alternating series usually looks like this:

$$\sum_{n=0}^{\infty} (-1)^n b_n$$

a big sign is the $(-1)^n$ we can test this series by seeing if the $\lim_{n \rightarrow \infty} b_n = 0$ and is decreasing by taking a derivative if we do this and we get zero and its decreasing its converging else its diverging

$$\#2) \sum_{n=1}^{\infty} \frac{8^n}{n} x^n \Rightarrow \lim_{n \rightarrow \infty} \left| \frac{8^{n+1} x^{n+1}}{n} \cdot \frac{n}{x^n 8^n} \right|$$

$$\lim_{n \rightarrow \infty} \left| \frac{\cancel{8^n} \cdot 8 \cdot \cancel{x^n} \cdot x}{\cancel{n} \cdot \cancel{x^n} \cdot \cancel{8^n}} \right| \rightarrow \lim_{n \rightarrow \infty} 8x \Rightarrow |8x| < 1$$

$$|x| < \frac{1}{8} \text{ this is } R, R = 1/8$$

$$-1/8 < x < 1/8$$

$$@ x = -1/8 \rightarrow \sum_{n=1}^{\infty} \frac{8^n}{n} (-1/8)^n \Rightarrow \sum_{n=1}^{\infty} \frac{\cancel{8^n}}{n} \frac{(-1)^n}{\cancel{(8)^n}} \Rightarrow \sum_{n=1}^{\infty} \frac{(-1)^n}{n}$$

this is a alternating series because it has a b_n which is $1/n$ and a $(-1)^n$ so:

$$b_n = \frac{1}{n} \rightarrow \lim_{n \rightarrow \infty} \frac{1}{n} = 0$$

$$\frac{d}{dn} (1/n) = -\frac{1}{n^2}$$

the limit is equal to zero and is decreasing so this is **converging**

$$② x = 1/8 \rightarrow \sum_{n=1}^{\infty} \frac{8^n}{n} (1/8)^n \Rightarrow \sum_{n=1}^{\infty} \frac{\cancel{8^n}}{n} \frac{(1)^n}{(\cancel{8})^n} \Rightarrow \sum_{n=1}^{\infty} \frac{1}{n}$$

this is a harmonic series so it will always **diverge**

$$\text{interval} = \left[-1/8, 1/8\right)$$

$$R = 1/8$$

$$\#3) \sum_{k=1}^{\infty} \frac{(-1)^k}{10^k} (x-3)^k \rightarrow \lim_{k \rightarrow \infty} \left| \frac{(-1)^{k+1} (x-3)^{k+1}}{10^{k+1}} \cdot \frac{10^k}{(-1)^k (x-3)^k} \right|$$

$$\lim_{k \rightarrow \infty} \left| \frac{\cancel{(-1)^k} (-1)^1 \cancel{(x-3)^k} (x-3)}{\cancel{10^k} 10^1} \cdot \frac{\cancel{10^k}}{\cancel{(-1)^k} \cancel{(x-3)^k}} \right| = \lim_{k \rightarrow \infty} \left| \frac{-(x-3)}{10} \right|$$

$$\left| \frac{(x-3)}{10} \right| < 1 \rightarrow |x-3| < 10 \quad \text{this is your } R, R=10$$

$$-10 < |x-3| < 10 \rightarrow -7 < x < 13$$

$$③ x = -7 \rightarrow \sum_{k=1}^{\infty} \frac{(-1)^k}{\cancel{10^k}} \cancel{(-10)^k} \Rightarrow \sum_{k=1}^{\infty} (1)^k$$

• this is a divergence series because it looks like $\sum a_n$

$$\lim_{k \rightarrow \infty} (1)^k = 1 \quad \text{so this diverges}$$

divergence series

the divergence series only test if it diverges a divergence series looks like this:

$$\sum_{n=0}^{\infty} a_n$$

if you take this limit $\lim_{n \rightarrow \infty} a_n$ and get non-zero it diverges but if you get zero its unknown

$\lim_{n \rightarrow \infty} a_n \neq 0$ diverges,

$\lim_{n \rightarrow \infty} a_n = 0 \rightarrow$ unknown

@ $x=13 \rightarrow \sum_{n=1}^{\infty} \frac{(-1)^n}{10^n} \Rightarrow \sum_{n=1}^{\infty} (-1)^n$

- this is a alternating series because of $(-1)^n$

$$b_n = 1, \quad \lim_{n \rightarrow \infty} 1 = 1$$

$$\frac{d}{dn}(b_n) = \frac{d}{dn}(1) = 0$$

• the limit is not equal to zero and its not decreasing so its diverging

$$\text{interval} = (-7, 13)$$

$$R = 10$$

up to what we have done right now is Part-1 we will go into Part-2 which is fairly simple

$$\sum_{n=1}^{\infty} n c_n x^{n-1} - \sum_{n=0}^{\infty} 7 c_n x^n$$

$$k = n-1$$

$$n = k+1$$

$$k = n$$

now find "k" using the equal "k=n-1" and replace all the n's with k

$$k = 1-1$$

$$k = 0$$

$$k = 0$$

$$\sum_{k=0}^{\infty} (k+1) c_{k+1} x^k + \sum_{k=0}^{\infty} 7 c_k x^k$$

we can factor our " $\sum_{k=0}^{\infty}$ " and " x^k " to get our final answer

$$\sum_{k=0}^{\infty} ([(k+1) c_{k+1} - 7 c_k] x^k)$$

the goal here is to get the summations to be the same then factor them out

$$\sum_{n=1}^{\infty} 8n c_n x^{n-1} + \sum_{n=0}^{\infty} 9c_n x^{n+1}$$

$$k = n-1$$

$$n = k+1$$

$$k = n+1$$

$$n = k-1$$

$$\sum_{k=0}^{\infty} 8(k+1) c_{k+1} x^k + \sum_{k=1}^{\infty} 9c_{k-1} x^k$$

We begin to see a problem both summations are not the same so we will pick the summation that has the larger number to be the one we want the other summations to match

$$\begin{aligned} \text{if } k=0 \\ \Rightarrow 8(0+1) c_1 x^0 \\ \Rightarrow 8c_1 \end{aligned}$$

• we picked $k=0$ because the next number is already one and that's the one we want

$$8c_1 + \sum_{k=1}^{\infty} 8(k+1) c_{k+1} x^k + \sum_{k=1}^{\infty} 9c_{k-1} x^k$$

↑
if $k=0$ we got
"8c₁" then if $k=1$
we get this

$$8c_1 + \sum_{k=1}^{\infty} ([8(k+1) c_{k+1} + 9c_{k-1}] x^k)$$

$$\sum_{n=2}^{\infty} n(n-1) c_n x^n + 2 \sum_{n=2}^{\infty} n(n-1) c_n x^{n-2} + 3 \sum_{n=1}^{\infty} n c_n x^n$$

$$k=n$$

$$k=n-2$$

$$n=k+2$$

$$k=n$$

$$\sum_{k=2}^{\infty} k(k-1) c_k x^k + 2 \sum_{k=0}^{\infty} (k+2)(k+1) c_{k+2} x^k + 3 \sum_{k=1}^{\infty} k c_k x^k$$

• we want the other summations to match this one

if $k=0$

$$\Rightarrow 2(2)(1) c_2 x^0 = 4 c_2$$

if $k=1$

$$\Rightarrow 2(3)(2) c_3 x^1 = 12 c_3 x$$

if $k=1$

$$\Rightarrow 3(1) c_1 x^1 = 3 c_1 x$$

• we do $k=1$ because we want the summation to be x

• we don't go to $k=2$ because $k=2$ is what we want the summation to be

$$\sum_{k=2}^{\infty} k(k-1) c_k x^k + 4 c_2 + 12 c_3 x + 2 \sum_{k=2}^{\infty} (k+2)(k+1) c_{k+2} x^k + 3 c_1 x + 3 \sum_{k=2}^{\infty} k c_k x^k$$

$$4 c_2 + 12 c_3 x + 3 c_1 x + \sum_{k=2}^{\infty} k(k-1) c_k x^k + 2 \sum_{k=2}^{\infty} (k+2)(k+1) c_{k+2} x^k + 3 \sum_{k=2}^{\infty} k c_k x^k$$

$$4 c_2 + 12 c_3 x + 3 c_1 x + \sum_{k=2}^{\infty} \left([k(k-1) c_k + 2(k+2)(k+1) c_{k+2} + 3k c_k] x^k \right)$$